Startup Intelligence AI

Indian Startup Funding Analytics, Machine Learning & Explainable AI

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1. Dataset

3,044 Indian startup funding events, January 2015 – January 2020. The raw CSV is never modified; cleaning writes a separate processed copy and a SQLite analytics database. Amounts and dates are parsed defensively, cities are normalized (Bangalore→Bengaluru, Gurgaon→Gurugram), and duplicates are flagged rather than dropped.

Year	Events	Recorded total
2015	933	$8.60B
2016	993	$3.83B
2017	687	$10.43B
2018	309	$5.12B
2019	111	$9.69B
2020	7	$390.21M

Sector	Events	Recorded total
E-Commerce	204	$8.25B
Consumer Internet	589	$6.25B
Transportation	4	$3.92B
Technology	310	$2.23B
Finance	57	$1.97B

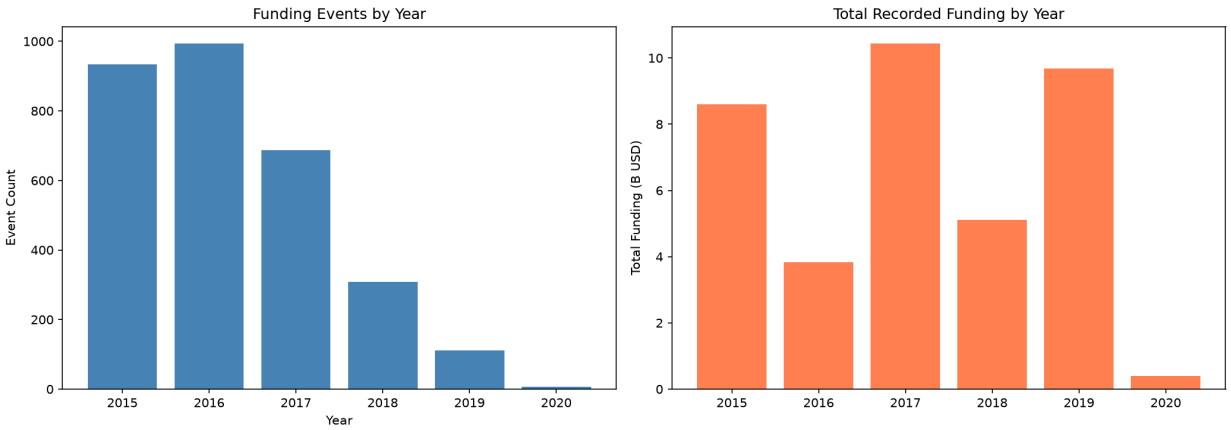


Figure 1 — Funding events and recorded total by year.

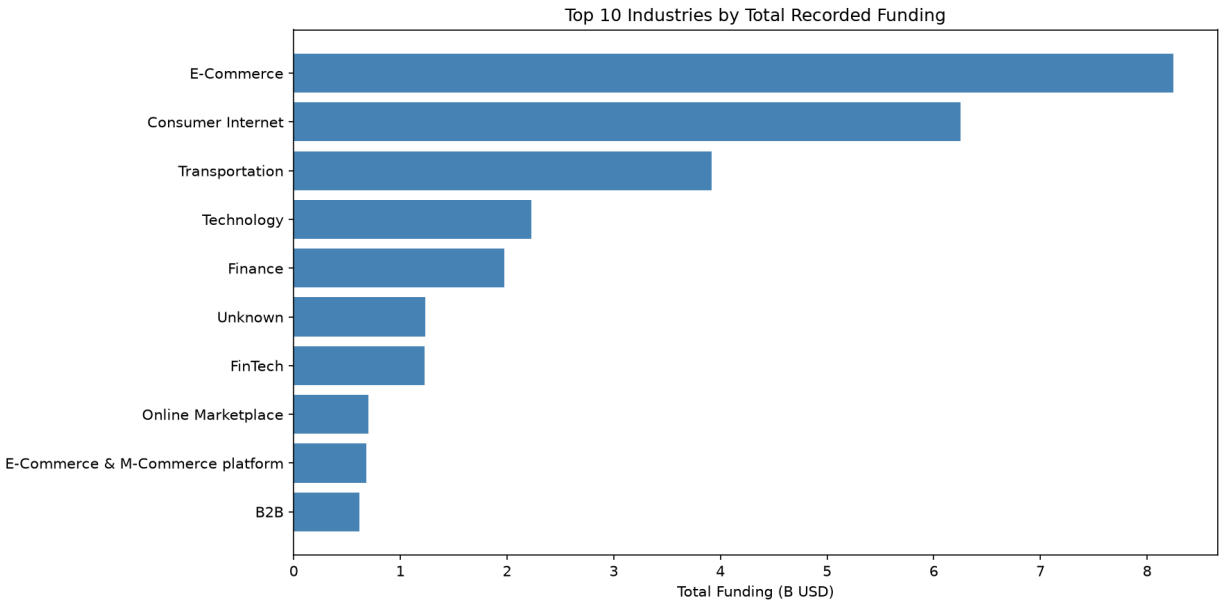


Figure 2 — Top sectors by recorded funding.

2. Modelling Approach

The target is extremely right-skewed (skew ≈ 22; the top 1% of rounds hold roughly 44% of all recorded funding). Raw-dollar metrics are therefore dominated by a handful of mega-rounds, so **RMSLE** is used as the selection metric: it measures relative error and is not hostage to outliers.

Design decisions, each verified by measurement:

• **Log target.** Every regressor is wrapped in TransformedTargetRegressor(log1p, expm1) so it trains in log space and predicts in raw dollars.

• **Group-median target encoding.** Median funding per industry×year, city×year, industry, city, investment type and industry×city is computed during fit on training rows only. Groups under 10 rows fall back to the global training median. *subvertical* is excluded: with ~1,300 distinct values across ~2,000 rows it degenerates into memorised one-offs.

• **Temporal validation.** 80/20 time-based split (train = earlier events). Hyperparameters are chosen by inner TimeSeriesSplit CV (4 folds) on the training period only. The winner is selected by inner CV score — never by test performance, which would invalidate the reported test number.

• **Baselines.** A global median dummy and a sector×year median are both reported, so improvement is measured against a strong comparator rather than a trivial one.

• **Investor features.** Implemented and tested but disabled by default: they measurably degraded RMSLE (RF 1.8686 → 1.9304). The code is retained so the negative result stays reproducible.

3. Regression Results

Model	CV RMSLE	Test RMSLE	MAE	R²
Random Forest (selected)	1.2741	1.9138	$35.44M	-0.0274
HistGradientBoosting	1.2752	1.9039	$35.42M	-0.0267
Ridge Regression	1.2759	2.1389	$35.36M	-0.0275
Baseline (Sector x Year Median)	—	2.4104	$36.63M	-0.0310
Dummy (Median)	—	2.4281	$36.69M	-0.0310

Selected hyperparameters: max_depth=20, **min_samples_leaf**=20, **n_estimators**=200

Every trained model beats both baselines on RMSLE. R² stays negative because squared error on raw dollars is still governed by outlier rounds — a property of the target, not evidence the model learned nothing. Round size is driven by traction, investor syndicate and timing, none of which appear in this dataset.

Note the CV/test gap (≈1.27 vs ≈1.90): funding drifts strongly upward over time, so 2015–2017 is genuinely harder than 2018–2020. Inner CV scores should be read as relative rankings, not absolute accuracy.

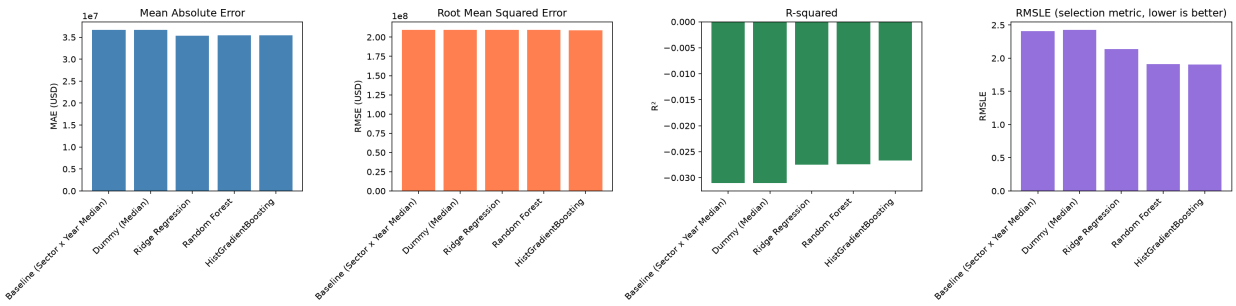


Figure 3 — MAE, RMSE, R² and RMSLE across all models.

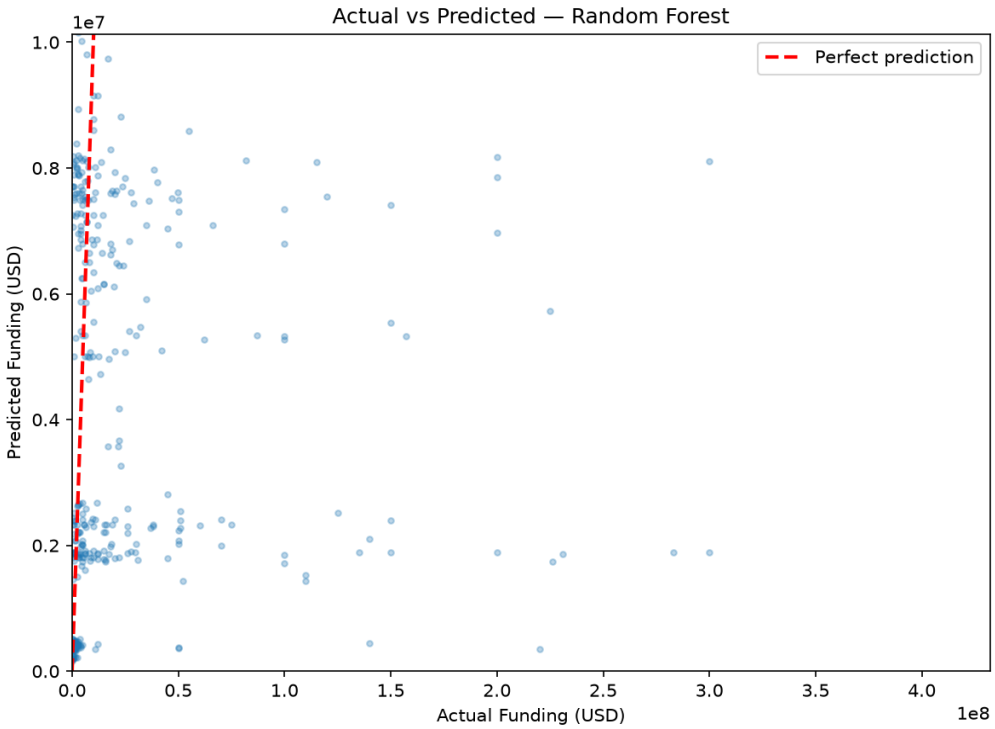


Figure 4 — Actual vs predicted, selected model.

4. Funding-Range Classification

Funding is binned into Low / Medium / High using percentile thresholds derived from **training** data only, then applied unchanged to the test period. Re-deriving cut points on the test set would leak distributional information across the temporal split — it changed the assigned category on 175 of 413 test rows.

Model	CV F1	Test F1
Logistic Regression (selected)	0.5317	0.5386
Random Forest	0.4833	0.5867
Dummy (Most Frequent)	—	0.1528

Tuned: **C**=0.1, **max_iter**=1000

Classification consumes the same feature pipeline as regression (via build_feature_pipeline), differing only in numeric scaling.

5. Clustering

K-means over sector, city, investment type, log1p(amount) and year. Cluster count is chosen by silhouette score rather than hard-coded. Amounts are log-transformed before clustering — with skew ≈ 22, raw dollars would make every cluster a pure amount tiering — and categoricals are one-hot encoded, since label encoding imposes a false ordering on names like Bengaluru vs Mumbai.

k = 3, silhouette **0.2182**, over 2,062 events.

Cluster	Events	Median amount	Avg year	Top industry	Top city
0	631	$5.15M	2015.4	Consumer Internet	Bengaluru
1	749	$310.0K	2015.8	Consumer Internet	Bengaluru
2	682	$5.00M	2017.7	Consumer Internet	Bengaluru

Clusters group historical funding events with similar characteristics. They are data-driven groupings, not real-world startup categories, and carry no forward-looking guarantee.

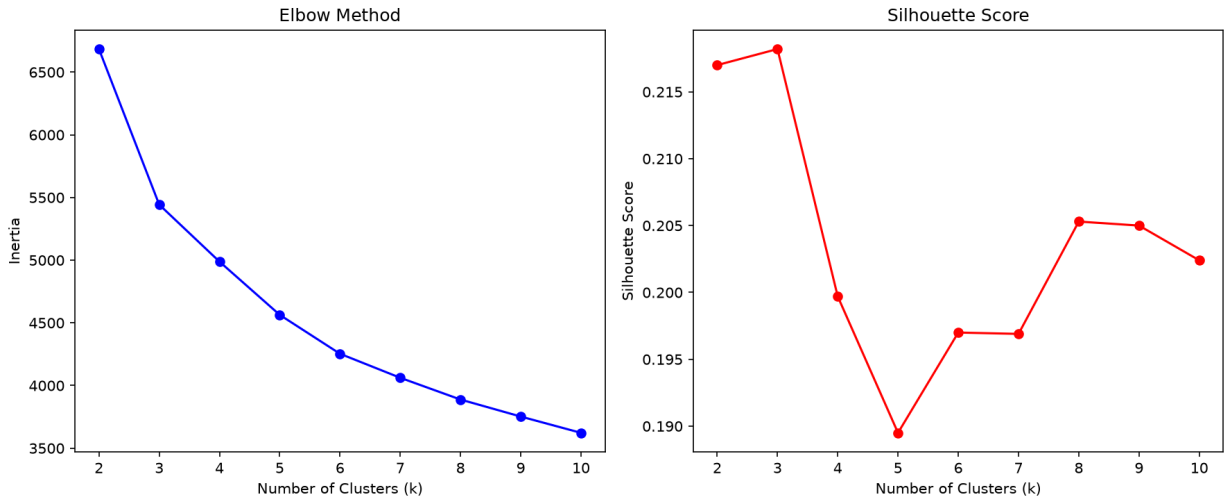


Figure 5 — Elbow and silhouette by k.



Figure 6 — Cluster composition.

6. Explainability

Permutation importance measures how much model error increases when an input column is randomly shuffled, so it covers the raw input columns only. The grp_* median-encoding features are generated downstream by GroupMedianEncoder and appear in the SHAP plot instead.

SHAP explains the transformed feature matrix. Because regressors train in log1p space, SHAP values are contributions to log amount rather than raw USD: a positive contribution means a larger predicted round, but not by a proportional dollar amount. SHAP is attempted opportunistically and degrades gracefully to permutation importance if unavailable. Feature importance reflects association, not causation.

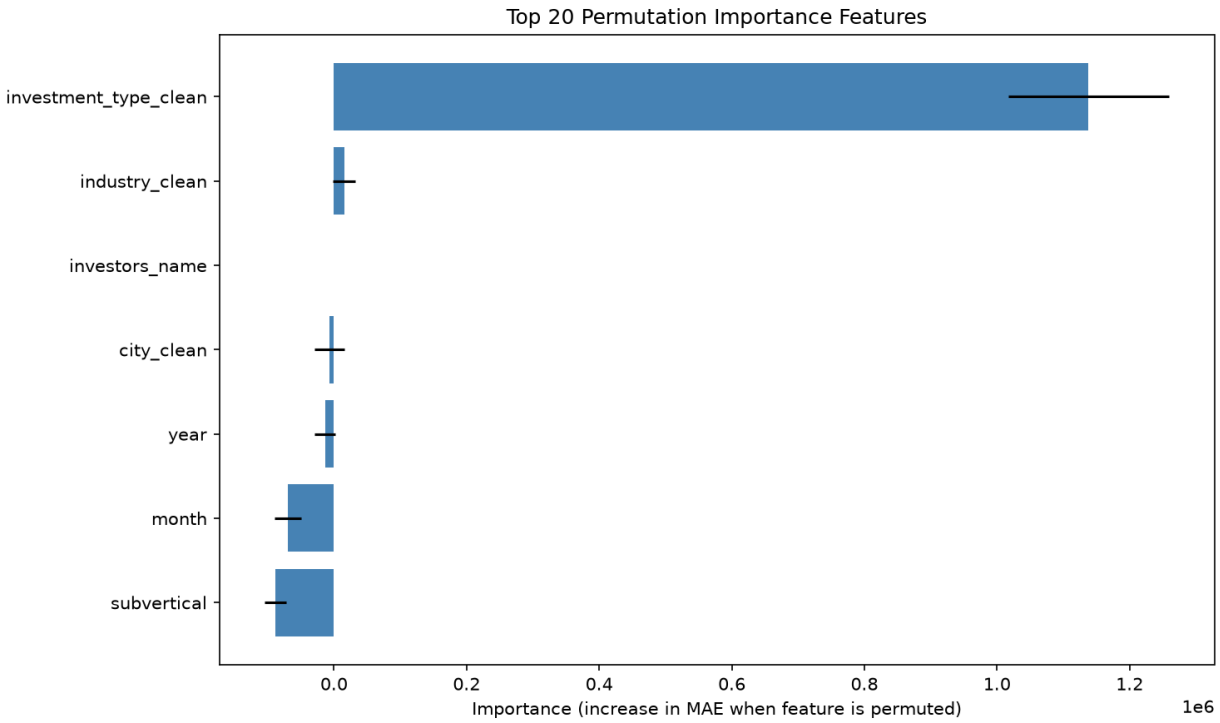


Figure 7 — Permutation importance.

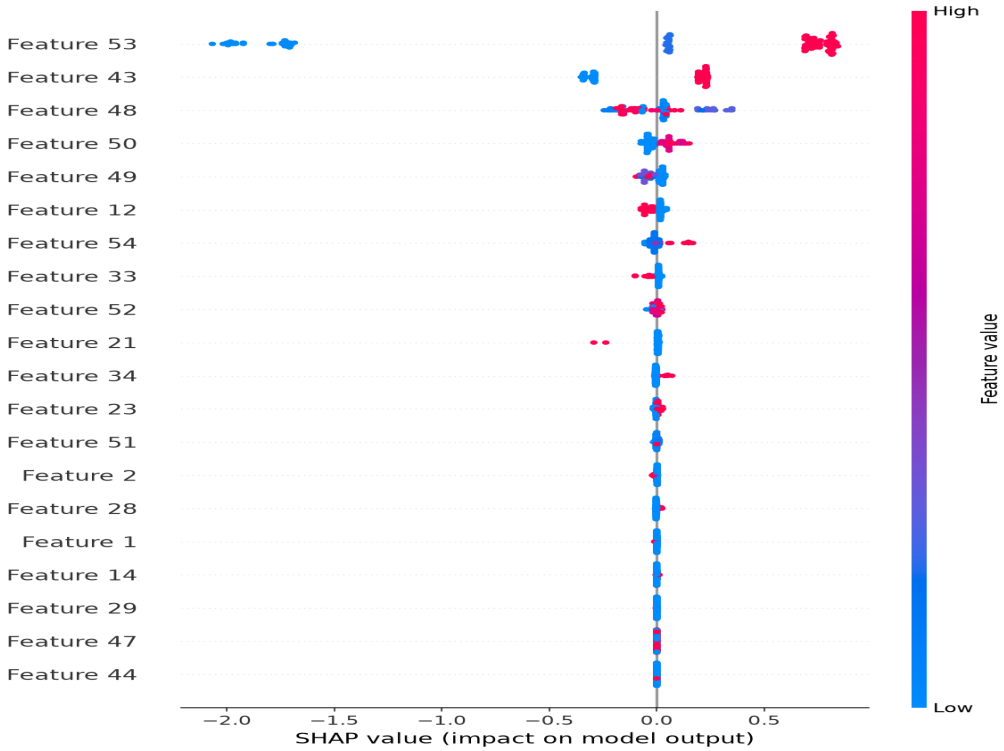


Figure 8 — SHAP summary (if available).

7. Limitations

• Historical records from 2015–2020 do not represent the current market.

• Missing funding amounts cut the modelling sample from 3,044 to 2,062 rows.

• R^2 is negative for every model; RMSLE is the metric to judge results by.

• Round size depends on traction, timing and syndicate quality — none of which are in this dataset. Predictions are exploratory estimates, not financial advice.

• Feature importance does not imply causation.

• The dataset contains outliers and inconsistent values that may affect analysis.

• CV scores are optimistic relative to the test period due to upward funding drift.